Emiliano Hansen

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EDUCATION

California Polytechnic State University **San Luis Obispo, CA**
B.S. in Mechanical Engineering *Expected Mar 2026*
GPA: 3.80 | Coursework: Composite Manufacturing, Vibrations, MATLAB, Heat Transfer, Mechanics of Materials

WORK EXPERIENCE

Apple – Vision Products Group **Sunnyvale, CA**
Product Design Engineering Intern *Mar 2025 – Sep 2025*
- Reduced system weight by 40% by using nTop optimization and replacing fasteners with laser welding
- Led reliability testing by designing components to withstand structural failures, improving durability by 25%
- Designed and tested ergonomic prototypes with varying configurations to evaluate optimal input placement

Formlabs **Boston, MA**
R&D Engineering Intern *Sep 2023 – Dec 2023*
- Prototyped mechanisms to significantly enhance powder flow efficiency by 60%
- Designed and manufactured a jig for modifying Alumina-4N resin wipers improving set up time by 8x
- Designed and produced jigs using Onshape to test products on manufacturing line

Delta Star **San Carlos, CA**
Mechanical Engineering Intern *Jun 2023 – Aug 2023*
- Completed study on plasma cutter to optimize material utilization saving an average of $130,000 annually
- Optimized inventory in machine shop to improve machine loading time by 50%

RESEARCH EXPERIENCE

Cal Poly Tensegrity Research Group – Soft Robotics & Tensegrity Structures **San Luis Obispo, CA**
Research Assistant *Mar 2023 – Present*
- Improved stiffness of soft robotic arm by 20% by making a mechanism to reorient forces placed on the arm
- Optimized weight of soft robotic arm by 30% by using nTop to create lattice structures using SLA and SLS
- Developed and refined an upper-limb tensegrity exoskeleton prototype to improve human functional support

Aizenberg Lab – Harvard SEAS REU **Cambridge, MA**
Research Assistant *Jun 2024 – Aug 2024*
- Developed steady-state heat flux experiment to test dynamic thermal conductivity of fluidic window prototypes
- Measured ~50% energy savings over common windows via radiation experiments using heaters and cold plates
- Used a Nanoscribe 3D micro-printer to fabricate high-resolution molds for micro-actuator prototypes

Cal Poly HVAC Lab – Atmospheric Water Harvesting **San Luis Obispo, CA**
Research Assistant *Nov 2022 – Jun 2024*
- Research focuses on an energy-efficient device to collect and condense water from humid air into drinking water
- Improved data collection methods by 10% by designing an environmental chamber to control humidity levels

EXTRACURRICULARS

NASA AMES Robotics, FRC & VEX Team 254 **San Jose, CA**
Manufacturing Lead & Design Member *Aug 2018 – Jun 2022*
- Designed tooling and jigs to improve throughput on high-volume parts, decreasing cycle time by 40%
- Redesigned the 2022 FRC robot to decrease weight by 10% while maintaining assembly integrity

AWARDS & HONORS

Columbia EngAGE Program Feb 2025
Discover ME Program – UT Austin Oct 2024
Formlabs In-Kind Sponsorship (Access Initiative) – Valued at $5K Jun 2024
Cal Poly CENG Student Grant – $1.5K Apr 2024
VEX Robotics Competition World Championship Winner Apr 2022
FIRST Robotics Competition World Championship Winner Mar 2022
FIRST Robotics Competition World Championship Finalist Mar 2019

SKILLS & CERTIFICATES

Technology: SolidWorks, Autodesk Inventor, Siemens NX, Onshape, nTop, KeyShot, Adobe Illustrator, Premiere Pro, Trello
Certificate: Project, Program, and Portfolio Management – ISO 21500 and ISO 21502 Certified